

Reg No.: _____

Name: _____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

Second Semester M.Tech Degree (R, S) Examination May 2024 (2022 Scheme)

Discipline: COMPUTER SCIENCE AND ENGINEERING**Course Code & Name: 222TCS002 DEEP LEARNING**

Max. Marks: 60

Duration: 2.5 Hours

PART A*Answer all questions. Each question carries 5 marks*

Marks

- 1 a) Explain the fundamental differences between deep learning and traditional machine learning approaches. - (2.5)
- b) Provide examples of tasks where deep learning excels over traditional methods. (2.5)
- 2 a) Recall Back Propagation in Convolution Neural Network (2)
- b) Discuss the trade-offs between max-pooling and average-pooling in CNN architectures - (3)
- 3 a) Sketch the architecture of an LSTM (Long Short-Term Memory) network. (2.5)
- b) Critically evaluate the challenges of vanishing and exploding gradients in RNN training and propose solutions. (2.5)
- 4 a) Analyze the advantages and limitations of word embeddings in capturing semantic meaning. (2.5)
- b) Evaluate the effectiveness of contextual word embeddings like BERT in capturing word semantics compared to static embeddings. (2.5)
- 5 a) Apply the concept of self-attention to explain how transformers capture contextual information in sequences. (2.5)
- b) Evaluate the effectiveness of multi-head attention in handling long-range dependencies in transformer-based models. (2.5)

PART B

Answer any 5 questions. Each question carries 7 marks

- 6 Consider a simple linear regression problem where you have a dataset with 50 data points. You want to use gradient descent to minimize the mean squared error (MSE) loss function. Initially, you choose a learning rate of 0.01 and randomly initialize the slope and intercept of the linear model. After running gradient descent for 100 iterations, you notice that the loss function seems to fluctuate without converging (7)
1. Evaluate the possible reasons for the lack of convergence in gradient descent.
 2. Suggest at least two strategies to address this issue and improve convergence.
- 7 a) Understand the architecture of residual networks (ResNet) and discuss their advantages over traditional CNNs (3)
- b) Given an input image with dimensions $32 \times 32 \times 3$, a convolutional layer with a kernel size of 5×5 , a stride of 1, and no padding, calculate the output volume size. Justify any assumptions made during the computation and provide a step-by-step solution, including the final output size. Your response should demonstrate a solid grasp of the mathematical operations involved in CNNs and their practical application. (4)
- 8 Apply the encoder-decoder framework to solve a machine translation task and explain the role of attention mechanisms (7)
- 9 a) Describe the process of creating word embeddings and their significance in natural language processing tasks (3)
- b) Given a vocabulary size of 10,000 words and a word embedding dimensionality of 300, calculate the total number of parameters required for training a Skip-gram model. Assume no additional parameters for the softmax layer or other model components (4)
- 10 Create a transfer learning pipeline for a computer vision task, including pre-training, fine-tuning, and model evaluation (7)
- 11 Evaluate the scalability of deep learning models in distributed computing environments and discuss potential bottlenecks. (7)

- 12 a) You are given a simple convolutional neural network (CNN) architecture for image classification. The CNN consists of two convolutional layers followed by max-pooling layers, and then two fully connected layers. The input image size is 32x32 pixels with three color channels (RGB). The first convolutional layer has 16 filters of size 3x3 with a stride of 1 and no padding. The second convolutional layer has 32 filters of size 3x3 with a stride of 1 and no padding. The max-pooling layers have a pool size of 2x2 with a stride of 2. The first fully connected layer has 128 neurons, and the second fully connected layer produces the final output. (4)

Calculate the total number of parameters in this CNN architecture

- b) Explain the dropout regularization technique in neural networks, (3)
